Luyang Zhao

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Research Interest

My current research spans a range of different topics in **Soft robots**, **Modular robots**, **Multi-robot Systems**, **Motion Planning**, **SLAM** and **Machine Learning**.

Education

Dartmouth College — **Hanover, NH**
Ph.D. Candidate in Computer Science — *Sep. 2018–Expected June 2024*
– Advisor: Prof. Devin Balkcom
– Honors: Neukom Outstanding Graduate Research Prize

University of Minnesota — **Minneapolis, MN**
B.S. in Computer Science — *Graduated **with distinction** in 2018*
– Research advisor: Prof. Maria Gini
– Honors: Dean's list

B.A. in Mathematics
– Senior project advisor: Prof. Karel Prikry
– Honors: Dean's list

Research Experience

Design of Soft Modular Robots — Dec. 2018 - July 2022
Dartmouth College (*Advisor: Prof. Devin Balkcom*) — *Hanover, NH*
- Led the development of **FlexBlocks** and **StarBlocks**, along with various soft modular robotic systems. Focused on designs for autonomous and collaborative tasks, aiming to **integrate modular and soft robotics**. For selected project demonstrations, please refer to the video links provided.: Link1 and Link2.

Motion Planning Research — Sep. 2018 - Nov. 2020
Dartmouth College (*Advisor: Prof. Devin Balkcom*) — *Hanover, NH*
- Conducted research in optimal robot motion strategies, including Bounds Planning and Piecewise Linear Regression Complex. Focused on developing **memory-efficient algorithms** and heuristic approaches for single and multi-robot motion planning.

Undergraduate Research — Sep. 2016 - May 2018
University of Minnesota (*Advisor: Prof. Maria Gini*) — *Minneapolis, MN*
- Developed an **assistive AI device** for individuals with memory loss, integrating voice-activated technologies for cognitive support; Designed an **object-finding robot** utilizing SLAM and advanced object detection algorithms for efficient navigation in unknown environments.

Work Experience

Applied Scientist Intern @ Amazon Robotics — June 2021 – Sep. 2021
Amazon — *Westborough, Massachusetts*
- Conducted in-depth research and development in the fields of **localization**, **path planning**, and **navigation** for **Amazon's autonomous mobility robots**. This involved identifying and addressing key challenges in robotic movement and efficiency.
- Designed and implemented innovative algorithms tailored to enhance robot performance. These solutions were effectively integrated into the robots' systems using **ROS** and **C++**, demonstrating a significant improvement in their operational capabilities.

Summer Research Intern @ TuSimple Planning — June 2022 – Sep. 2022
TuSimple — *San Diego, California*
- Developed an advanced **local turn lateral trajectory optimization** for **autonomous trucks**, utilizing a **kinematic tractor-trailer model**. This model enhanced the precision in describing the kinematic behavior during local turns, significantly improving upon the traditional point-mass model.
- Focused on achieving **obstacle avoidance** through the generation of optimized lateral trajectories. Employed **sequential quadratic programming** techniques to effectively minimize a cost function that included state cost, input cost, and jerkiness, while adhering to necessary constraints.
- Successfully implemented and integrated the optimized trajectory solution in a **real-time environment** using **ROS** and **C++**. Demonstrated improved efficiency and safety in autonomous truck maneuvers.

Workshops Co-Organized

Tensegrity Robotics Workshop at IROS 2023 – 2023 *Co-organized and spoke at this innovative workshop focusing on tensegrity robots. Presented on 'Untethered Self-Reconfiguring Morphable Modular Robots for Field Deployment,' highlighting advancements in robotic mobility and adaptability in unstructured environments.* [Link: https://www.eng.yale.edu/faboratory/tensegrityworkshop/]

Making Your Own Compliant Mechanisms for Soft Robots (Proposed for RoboSoft 2024) – 2024 *Submitted to speak and facilitate at this upcoming workshop, which delves into compliant mechanisms in soft robotics. The session aims to explore materials like carbon fiber/epoxy and metal in creating adaptable robotic designs, enhancing the functionality of soft robots in diverse environments. The workshop will combine theoretical knowledge with practical application, contributing to the innovation in soft robotics.*

Mentorship Experience

Lab Mentor Sep. 2018 – Now
Dartmouth Reality and Robotics Lab *Hanover, NH*

- Master students: Yijia Wu (2021-2022, now PhD student at WPI), Weishu Zhan (2022), Yitao Jiang (2022-now)
- Undergraduate students: Josiah Putman (now in Google), Maxine Perroni-Scharf (now PhD student at MIT)

Teaching Assistant Sep. 2018 – Now
Dartmouth College *Hanover, NH*

- **CS81/281** : Principles of Robot Design and Programming – 2018 Fall
- **CS76/276** : Artificial Intelligence – 2018 Winter, 2019 Fall and 2023 Fall
- **CS1** : Introduction to Programming and Computation – 2019 Spring and 2020 Spring
- **CS50** : Software Design and Implementation – 2019 Summer

Teaching Assistant, Summer Computing Academy June 2017
University of Minnesota *Minneaplis, MN*

- Assisted senior high school students in developing programs for Scribbler robots, image processing, video, 3D printing, and other applications.

Honors and Awards

2nd place in the **Neukom Outstanding Graduate Research Prize**, $750 (2023)
Undergraduate **Research** Opportunities Program (UROP) **Scholarship**, $1500 (2016)

Manuscripts in Preparation

[1] FlexBlocks: Untethered Self-Reconfiguring Morphable Modular Robots for Field Deployment (In preparation to submit to Science Robotics in Dec.)

[2] Unsupervised Reinforcement Learning for Task-Specific Design of Soft Modular Robots (In preparation to submit to RSS in Feb.)

[3] SeeSea: Multi-modal 3D Perception Dataset of In-water Obstacles for Navigation of Autonomous Surface Vehicles (In preparation to submit to IJRR in Dec.)

Publications

[1] **Luyang Zhao**, Y. Wu, W. Yan, W. Zhan, X. Huang, J. Booth, A. Mehta, K. Bekris, R. Kramer-Bottiglio, and D. Balkcom. Starblocks: Soft actuated self-connecting blocks for building deformable lattice structures. *IEEE Robotics and Automation Letters*, 8(8):4521–4528, 2023

[2] **Luyang Zhao**, Y. Wu, J. Blanchet, M. Perroni-Scharf, X. Huang, J. Booth, R. Kramer-Bottiglio, and D. Balkcom. Soft lattice modules that behave independently and collectively. *IEEE Robotics and Automation Letters*, 7(3):5942–5949, 2022

[3] **Luyang Zhao**, J. Putman, W. Wang, and D. J. Balkcom. PLRC*: a piecewise linear regression complex for approximating optimal robot motion. In *International Conference on Intelligent Robots and Systems, IROS 2020*

[4] J. Putman, L. Oh, **Luyang Zhao**, E. Honnold, G. Brown, W. Wang, and D. J. Balkcom. Piecewise linear regressions for approximating distance metrics. *CoRR*, abs/2002.12466, 2020

[5] J. Putman, L. Oh, **Luyang Zhao**, E. Honnold, G. Brown, W. Wang, and D. J. Balkcom. LLDM: locally linear distance maps for robot motion planning: Extended abstract. In *2019 International Symposium on Multi-Robot and Multi-Agent Systems, MRS 2019*

[6] L. Ferland, Z. Li, S. Sukhani, J. Zheng, **Luyang Zhao**, and M. L. Gini. Assistive AI for coping with memory loss. In *The Workshops of the The Thirty-Second AAAI Conference on Artificial Intelligence, 2018*, AAAI Workshops

Conference and Seminar Presentations

[1] "FlexBlocks: Untethered Self-Reconfiguring Morphable Modular Robots for Field Deployment" Presented in ICRA 2023 Tensegrity workshop

[2] "Soft Modular Robotics: Design and Applications" invited talk at Rutgers university robotics seminar (10/17/2022 10:30am-12:00am)

[3] "StarBlocks: Soft actuated self-connecting blocks for building deformable lattice structures." Presented in ICRA 2022 MSRR workshop and NERC2022

[4] "Soft Lattice Modules That Behave Independently and Collectively," Presented in Robosoft 2022

[5] "PLRC*: a piecewise linear regression complex for approximationg optimal robot motion", Presented in IROS 2020

[6] "LLDM: locally linear distance maps for robot motion planning", Presented in MRS 2019

References

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Prof. Xiaonan (Sean) Huang
Assistant Professor, Robotics
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Prof. Rebecca Kramer-Bottiglio
Associate Professor, Mechanical Engineering & Materials Science
Yale University Email: rebecca.kramer@yale.edu